

Data Mining Report

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April 28, 2026

Pre-processing

1.1 Combining Attributes and Numerical Data:

For low instance variables, a manual assignment from numerical values to worded categories was taken. However, in the cases of large variable assignments it was easier to merge attributes with the appropriate columns in the table and then prune undesired columns and re-name the new variable columns. For remaining values, Formula widget was used to populate values.

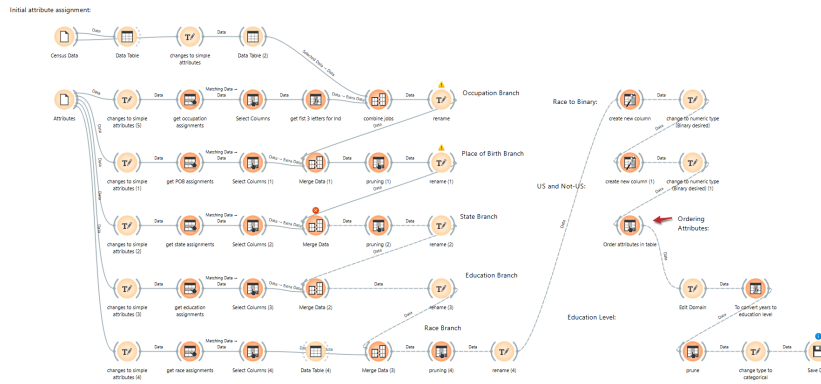


Figure 1.1: Node structure for turning numerical data into attribute values.

1.2 Outlier and Missing data:

Using a subset of 5000 records, the 'Outlier' widget was used to identify and categories outliers. While generally good practice, there could also exist trends within these extremes themselves; for example, all of these cases could have attended the same level of education.

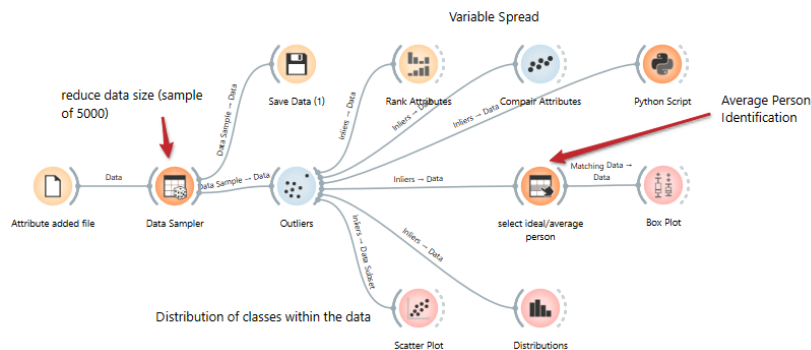


Figure 1.2: Identifying and Removing Outliers.

1.3 Dataset Overview & Summary Statistics:

Race With some minority groups, such as "Alaska Native", with only one representative no general prediction could be made. As a result, minority groups are combined to form a binary classification of **White** and **Not-White**. Similarly, **place of birth** was also culminated into **US** and **Non-US** to help reduce skewness and maintain data validity.

Frequency distributions necessitated combining groups for statistical robustness. While **White** participants composed 78% of the data, the next largest category was only 9%. Similarly, all non-US regions combined totaled only 16%. Consolidation ensures more reliable comparisons across categories.

The same is also said for **education**, with many peoples educational background varying largely based on how long they spent in school, it is easier to group by their achievements (checkpoints) within this field. Specifically, categories 1 $\rightarrow$ 15 were mapped to **no-diploma**, 16 $\rightarrow$ 17 to **high-school**, and 18 $\rightarrow$ 19 to **post-high-school**, with subsequent levels representing higher-tier qualifications.

Finally, the **Industry** attribute was extracted to identify broader economic trends. Using specific job titles would have introduced excessive "noise," as identical roles often have varied titles based on company or location. By grouping these into industrial sectors, the data reveals systemic trends rather than isolated instances.

1.3.1 Preliminary Observations

With the main aim being to predict a persons' **income**, it is good to establish a baseline to compare to. In this case, it consists of choosing someone that fits in to the most common category. Following from the data refinement, the most common person consists of someone:

- **White**
- **Married**
- Works in an **Office**
- Achieved qualifications **Post High-school**

, this person earns anywhere from \$21,000 $\rightarrow$ \$70,000. However, the mean income for someone with all of these attributes is \$62,881.25; way above the expected \$39,000. This large deviation is a result of positive skewness within the demographic, where a minority of high-earners pulls the average upward,

misrepresenting the typical individual. To mitigate this, the following analysis linearises the relationship between attributes and earning, allowing for a more accurate comparison.

Fairness in income distribution

2.1 Overall Distribution of Income:

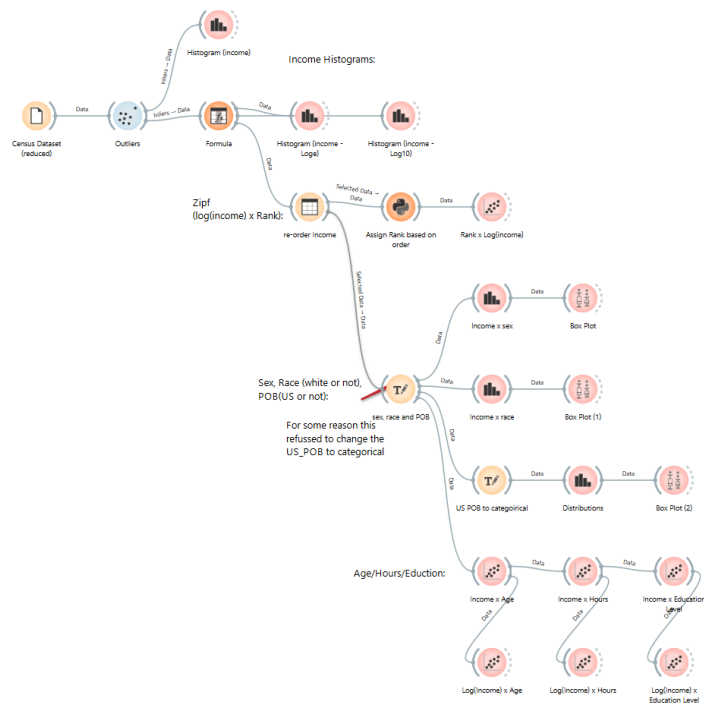
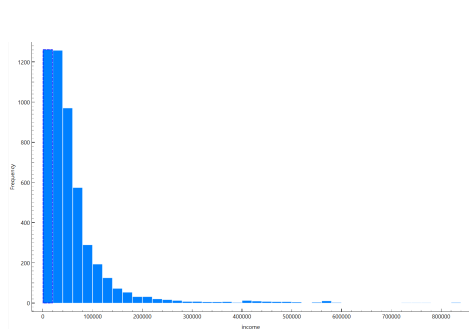
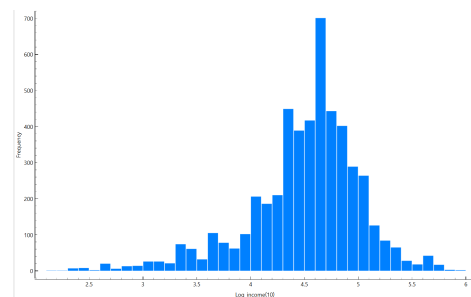


Figure 2.1: How the data was extracted.



(a) Raw Income Distribution.



(b) Income distribution using $\log_{10}$ scale.

Figure 2.2: Comparing income distribution metrics.

As seen in Figure 2.2a, the income distribution is more **reciprocal** in nature as shown by its positive skewness. This is a result of most individuals earning in the lower brackets, with higher paying jobs being distributed more sparsely. To linearise this distribution and break down the $1/x$ relationship, a $\log_{10}$ transformation is used. This base was chosen over the natural logarithm ($\ln$) because it provides a more intuitive scale for financial data.

2.1.1 Additional notice:

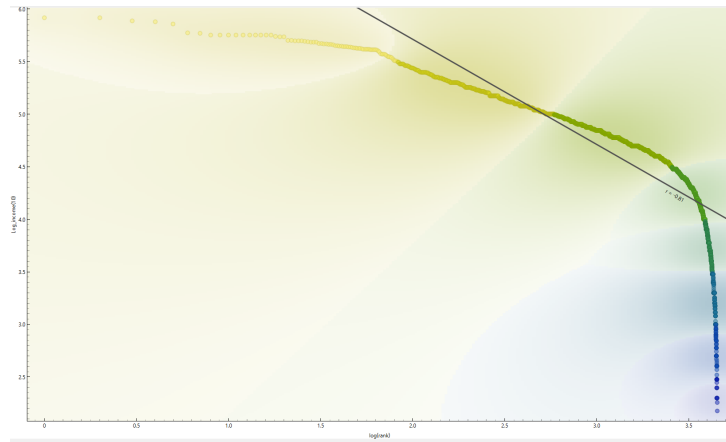
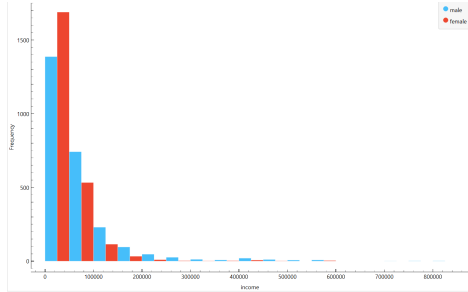


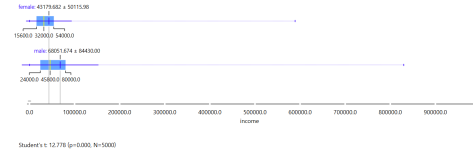
Figure 2.3: Log(income) vs log(income ranks)

Another way to see how wealth is distributed is shown through a Zipf Plot (Figure 2.3). The negative slope suggests that income drops off rapidly as rank increases, demonstrating a high concentration of wealth among top-tier individuals. The plateau at the lowest ranks (top-left) represents the highest earners, while the sharp descent toward the higher ranks illustrates how income diminishes for the majority of the population. The $r = -0.81$ correlation confirms a strong inverse relationship between rank and income, with the log-log representation confirming that the data follows a Pareto distribution (based on the idea that a small proportion of instances hold a lot of the value within the data - few people earn a lot of money).

2.2 Effect of Sex on Income distribution:



(a) Distribution of Sex with Income.

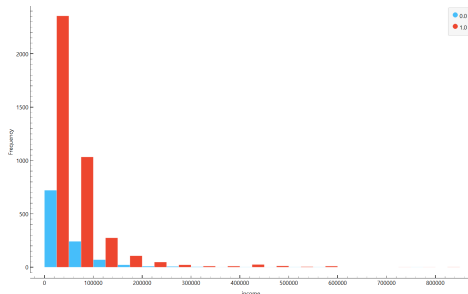


(b) How does gender affect the distribution of income.

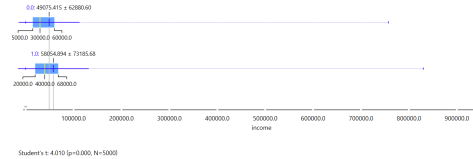
Figure 2.4: Effect Sex has on Income.

Not only are there visual differences between genders, where female representation declines more sharply than male representation as income increases, but there is also statistical evidence. A T -statistic of 12.778 suggests the disparity is not a fluke caused by sampling bias. This is supported by the P -value; since $P < 0.05$, there is sufficient evidence to reject the assumption of equality.

2.3 How does Race affect Income:



(a) Distribution of Race with Income.



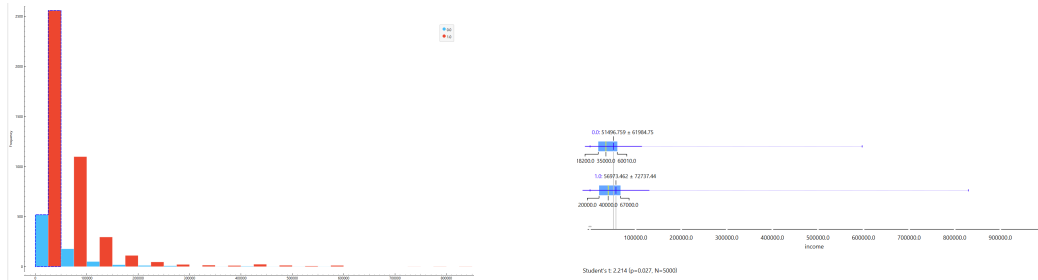
(b) Comparison of those who are White and those who are Not-White on average income

Figure 2.5: Effect of Race on Income (White=1, Other=0).

Across all groups there are more White representatives. With a median difference of \$10,000 and $P < 0.05$, there is sufficient evidence to suggest **race** is a significant predictor to income; but less than gender since the T -test (4.04) is lower. Furthermore, there is also a greater inequality (range in

income) within the **white** group, shown by the greater standard deviation ($\pm 73K$ vs $\pm 62K$).

2.4 Does Place of Birth have an effect on Income?



(a) Distribution of Income with those born in the US.

(b) How does being born outside the US affect your income.

Figure 2.6: Effect where you are born has on Income (US=1, Not-US=0).

Despite having the smallest t -statistic ($t = 2.214$), a median difference of \$5,000 and $P < 0.05$ still provide sufficient evidence to suggest that birth region significantly influences income. Furthermore, a \$10,750 difference in standard deviation suggests that those born in the US experience higher income volatility and a wider range of financial outcomes compared to those born elsewhere.

2.5 Personal and Professional Drivers of Income

2.5.1 Age

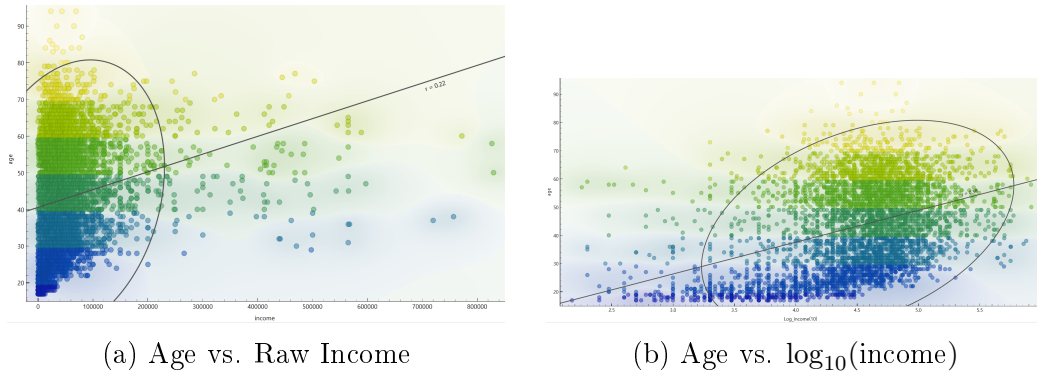


Figure 2.7: Correlation between Age and Income metrics.

The raw data in Figure 2.7a is difficult to interpret because the high-income outliers stretch the Y-axis, causing the majority of data points to pool at the bottom. The log-transformed version (Figure 2.7b) is significantly better as it linearises the growth trend. It suggests that as individuals age and gain experience, their earning potential increases, which is reflected in a more consistent R value and a clearer upward-sloping trend line.

2.5.2 Education

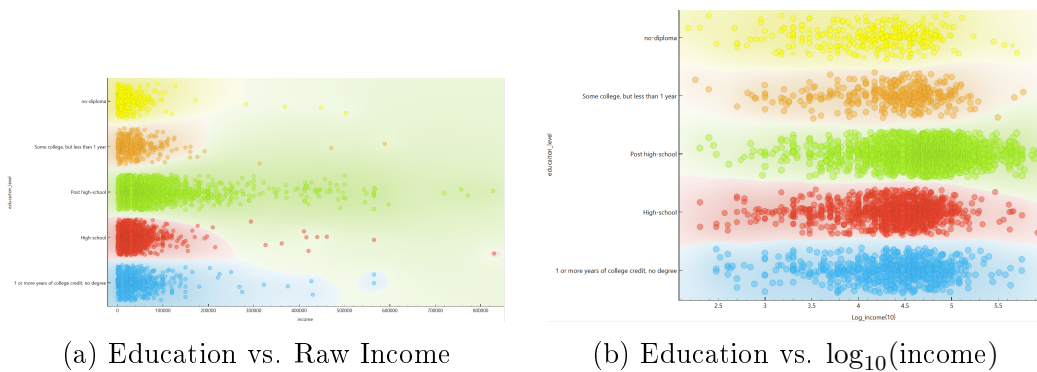


Figure 2.8: Correlation between Education level and Income metrics.

In the raw plot (Figure 2.8a), the extreme positive skew compresses the data, making central tendencies indistinguishable beyond the lack of high-earners

in the `no-diploma` group. Conversely, the $\log_{10}$ transformation (Figure 2.8b) normalises this variance. Showing visibly that as duration in education increases, so does the general income cluster.

2.5.3 Hours Worked

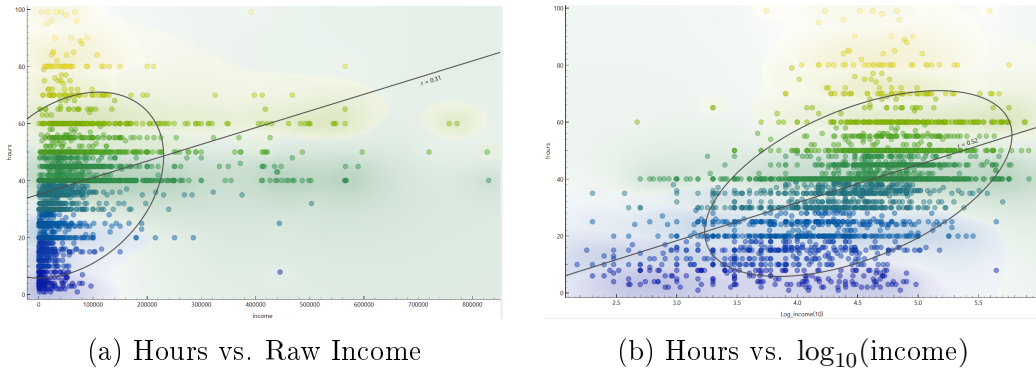


Figure 2.9: Correlation between Hours Worked and Income metrics.

While the relationship in Figure 2.9a is obscured by outliers the $\log_{10}$ transformation in Figure 2.9b reveals a clearer linear trend. A correlation of $R = 0.52$ indicates a moderate positive linear relationship indicating hours worked could be an appropriate predictor for income. The elliptical cluster also indicates that as hours increases, so does the income floor (for most cases), with widening variance at higher earning levels.

Predicting Income

3.1 Value of Education

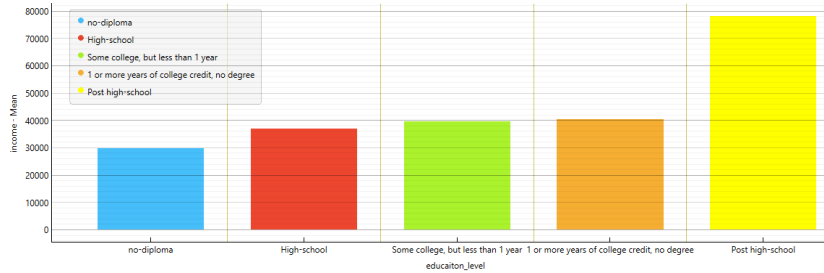


Figure 3.1: Average income with respect to education level

Table 3.1: Average Income by Education Level

Education Level	Avg. Income (\$)
No Diploma	30 068.50
High School	37 065.50
Some College (<1 year)	39 678.80
College (1+ years but no degree)	40 556.70
Post High-School	78 343.00

Further supporting 2.5.2, figure 3.1 illustrates a clear positive correlation between educational attainment and average income. Evaluating education on a year-by-year basis is problematic because time does not always equate to a standardised outcome or formal accreditation. Value can vary between years, with final years (when you get a certification) forming income spikes. Current jobs may also not be influenced by time in education due to experience or the fact that they may be overqualified.

3.2 Predicting Income Bracket

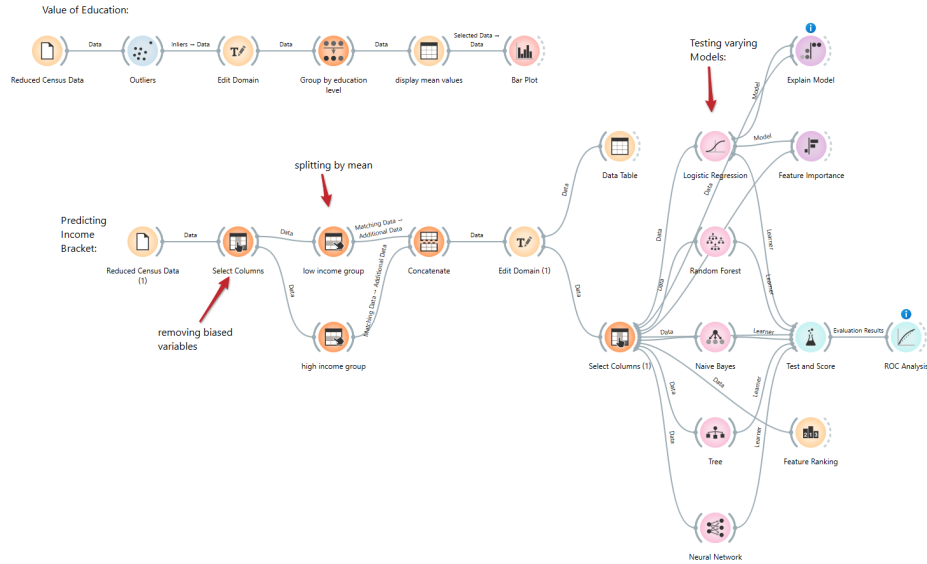


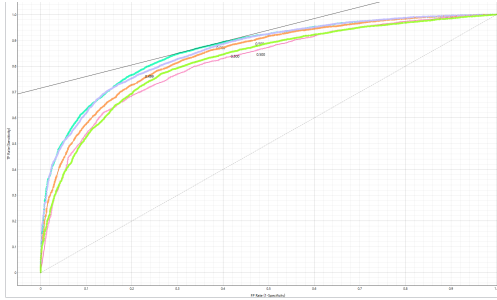
Figure 3.2: Model training and explainability structure.

After removing the **income** numerics to prevent data leakage, the following results were found:

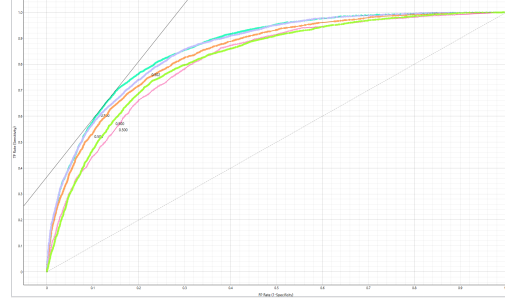
Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.864	0.802	0.798	0.797	0.802	0.523
Random Forest	0.843	0.787	0.777	0.779	0.787	0.474
Naive Bayes	0.860	0.774	0.781	0.798	0.774	0.519
Neural Network	0.818	0.770	0.767	0.765	0.770	0.451
Tree	0.813	0.758	0.754	0.752	0.758	0.420

Figure 3.3: Model comparisons on determining if someone's income will be high or low

Logistic Regression is the winning model with a **Classification Accuracy** of 80.2% and an **AUC** of 0.864. That a simple model outperforms more complex models (Neural Networks) suggests a strong linear correlation between the attributes and classification, providing a more generalised model that avoids overfitting.



(a) Predictively for Low Income bracket.



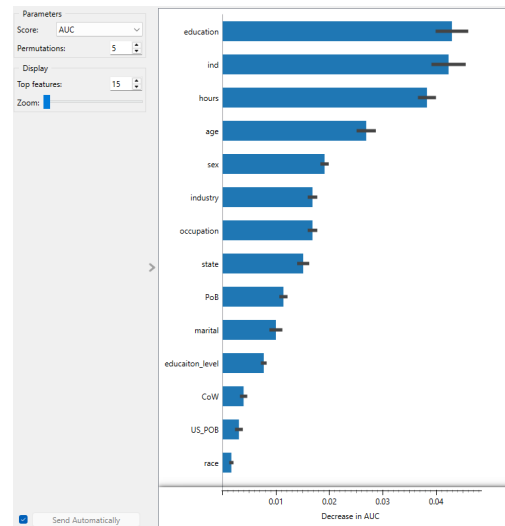
(b) Predictively for High Income bracket.

Figure 3.4: Logistics Regression outperforming for both classifications.

3.3 Feature Ranking

		#	Info. gain	Gain ratio	Gini	χ^2	ReliefF	FCBF
1	hours		0.096	0.054	0.051	376.055	0.016	0.078
2	education		0.101	0.052	0.060	491.881	0.011	0.077
3	education_level	5	0.083	0.042	0.048	14.693	0.044	0.000
4	occupation	529	0.310	0.041	NA	35689.489	0.152	NA
5	industry	804	0.310	0.041	NA	3046.029	0.152	NA
6	marital	5	0.056	0.036	0.031	676.043	0.058	0.048
7	ind	28	0.150	0.035	NA	58.761	0.170	NA
8	age	28.0	0.061	0.031	0.031	186.773	0.008	0.000
9	sex	2	0.025	0.025	0.014	78.660	0.032	0.000
10	race	804	0.014	0.012	NA	0.029	0.028	NA
11	PoB	804	0.051	0.009	NA	206.917	-0.012	NA
12	CoW	9	0.011	0.006	NA	18.102	0.004	NA
13	state	804	0.016	0.003	NA	143.544	-0.030	NA
14	is_white		0.001	0.001	0.001	1.278	0.022	0.000
15	US_POB		0.001	0.001	0.000	0.545	0.010	0.000

(a) Feature ranking for predicting income low/high



(b) Model-based feature importance

Figure 3.5: Comparison of data-driven feature ranking and model explainability.

Occupation and **industry** yield the highest Information Gain (0.310), indicating primary predictive power. However, the high granularity of **industry** suggests potential overfitting; **ind** (the third most influential feature) is used to reduce this effect. Conversely, demographic features like **US_Place-of-Birth** show negligible influence. The alignment between these rankings and Figure 3.5b confirms the model is transparent, making predictions based on

initial data rather than unintuitive correlations. However, **occupation** lost influence in the regression model despite its high initial rank; likely due to its **high cardinality**, creating noise.

3.4 Prediction Influence

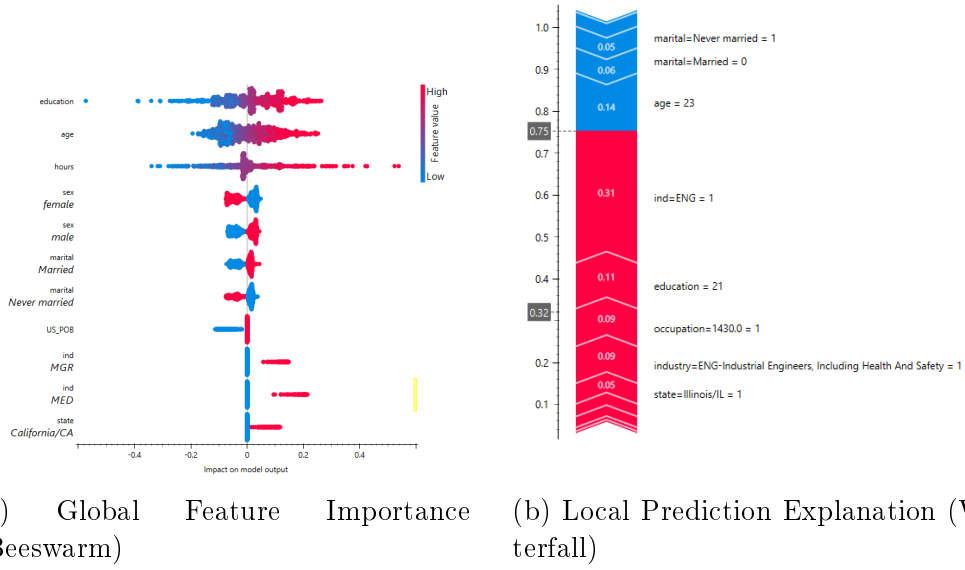


Figure 3.6: Model explainability using SHAP: (a) shows the overall impact of features across the dataset, while (b) illustrates the specific feature contributions for a single high-income prediction.

The SHAP summary plot (Figure 3.6a) illustrates the directional influence of key features. Higher values of education, age and working hours (red) increase the probability of a **high income** classification. Conversely, being **female** pushes the classification towards the **low income** (represented by blue). Industry sectors such as MGR(managerial) or MED(medical), also suggest a higher income, validating the importance of professional sector over geographical location. The SHAP waterfall plot (Figure 3.6b) provides a granular view on how features interact to a classification. This shows how those in the **Engineering** industry (+0.31) and **higher** education (+0.11) push the class to **high income** despite items like age pushing it down. Showing how professional and educational background successfully override the demographic disadvantage, puhsing the prediction above the 0.32 baseline.

Demographics of US Elections

4.1 Mapping Data

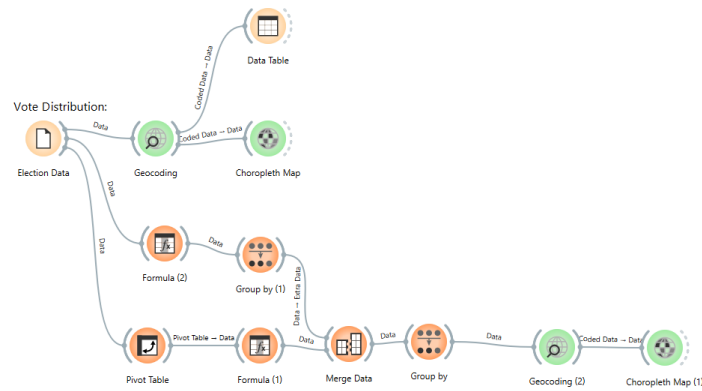


Figure 4.1: Expressing geographical data.

4.1.1 Election Results:

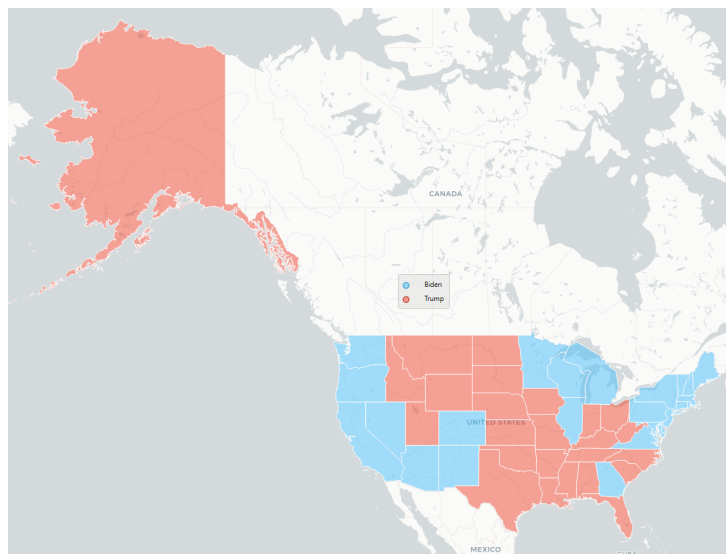
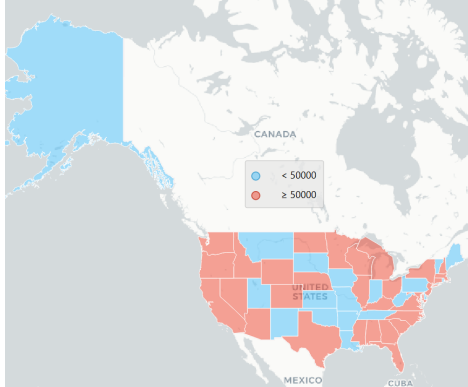


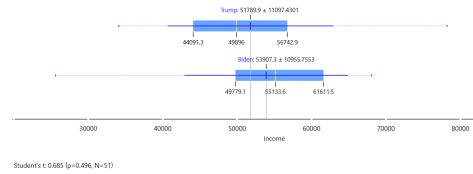
Figure 4.2: Winning Parties between states

Figure 4.2 shows which of the top running parties was dominant in which state in 2020.

Effect of Income on Voting habits:



(a) Raw income of voters on average (by state)

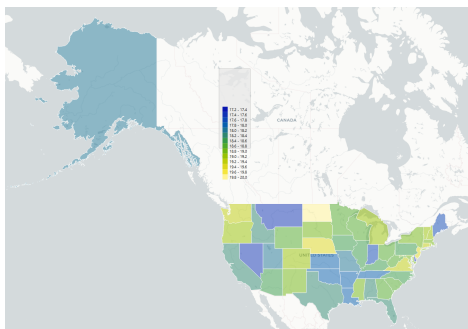


(b) is the average income high or low?

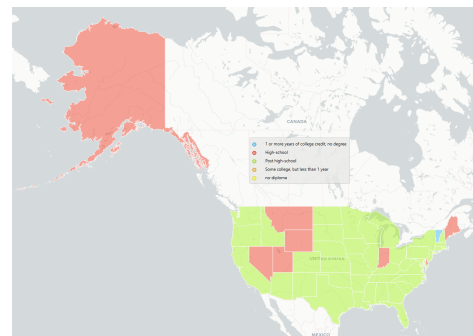
Figure 4.3: Income distribution between states

Initial High vs Low income state decision is not feasible because averaging out state income is diluted by extreme income which results in 50/51 states being high-income. This was replaced with a median data split which results in 41/51 states following the pattern of: Biden = High-Income and Trump = Low-Income. With a regression of ± 0.13 between raw income and state vote there is not enough evidence to suggest this is not a result of noise ($p = 0.496$).

Education Correlation in voting:



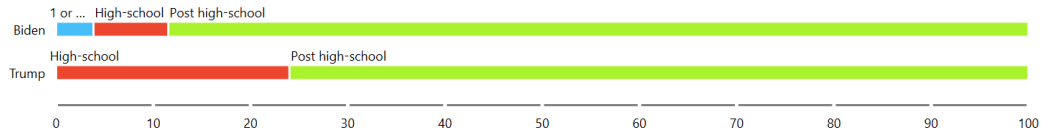
(a) Education numerical comparisons between states



(b) Average education level between states

Figure 4.4: Education distribution between states

With initial education distribution being unclear as to which candidate they voted for (after initial comparison), further analysis was required to determine if state-level averages masked underlying trends in voter attainment.



$\chi^2: 3.36$ ($p=0.186$, $\text{dof}=2$)

Figure 4.5: Distribution of average voters in states for Biden and Trump

Figure 4.5 shows how each candidate's voters were spread by taking a state-wide average of education level. While a **visible** reduction in education levels in states that voted more for Trump there is insufficient evidence that education level correlates to who they vote for (expressed by $P = 0.168$).

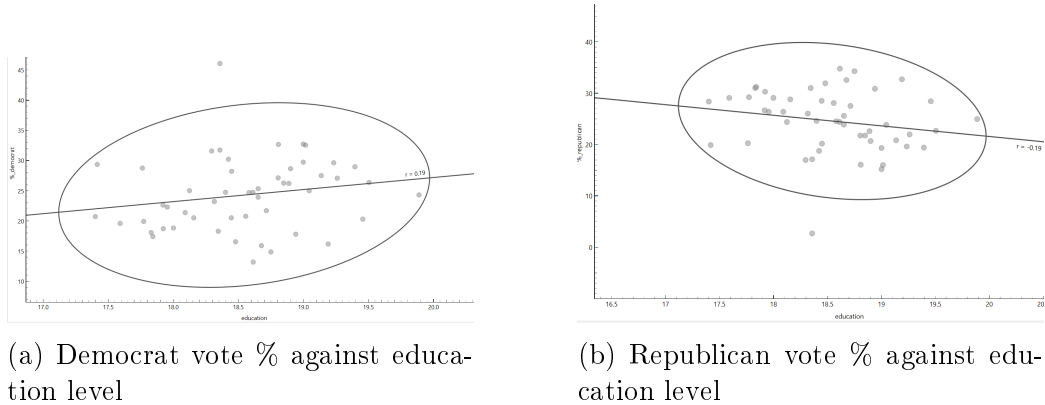


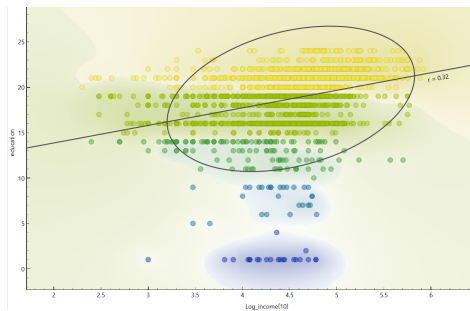
Figure 4.6: Education vs voting bias

By comparing the percentage distributions however, the effect of varying state size is removed. As a result, the graphs in figure 4.6 shows a weak correlation between the attributes which suggests it is not the primary driver of voting habits, rather an indication that those with lower income are more likely to vote Trump (but not exclusively).

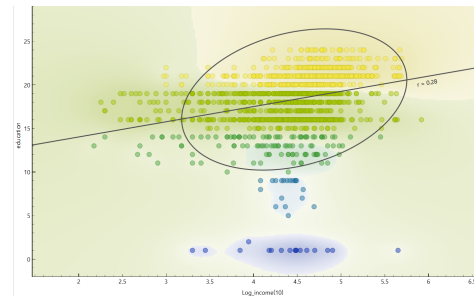
My Own Data Mining

Hypothesis: In Biden-voting states, income is more strongly correlated with education level, whereas in Trump-voting states, income is more strongly correlated with 'hours worked'. *Prove or disprove if there is evidence to suggest the wealth in the Democratic states is knowledge orientated (more white collar work), whereas in then states where Trump had more residence, the wealth was found in more blue collar workers.*

5.0.1 Underlying Trends



(a) Democrat voters education level vs their income.



(b) Republican voters education level vs their income.

Figure 5.1: Education vs Income in each subgroup of voters split by state winning party.

As previously discussed, there is a greater correlation between the Education level and Income in states which voted more for Biden, with $r = 0.32 > p = 0.28$. This helps to identify that in Biden voting states, Education relates more to their income, compared to those that voted Trump.

To further illustrate this point, the department/type of job that people get their income from will be checked.

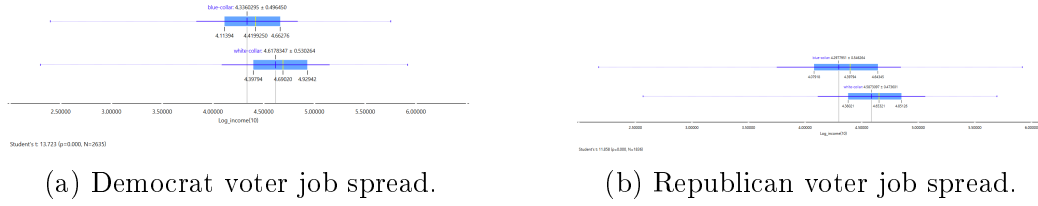


Figure 5.2: How are Blue and White Collar jobs distributed.

While the income for 'White Collar' jobs (requiring greater Education levels) remain the greatest across all states, those in Biden voting states appear to earn $\approx \$2,000$ more. With the White-collar distribution in Biden states having a higher mean income, compared to those in Trump voting states. However, with wider whiskers in Biden states, it suggests that there is a greater degree of income inequality compared to those in Trump biased states, where it shows a more middle-class distribution.

5.0.2 Predictive Voting

Limitations

Since voting statistics are state wide distributions and do not correlate to specific individuals, it is harder to determine who someone will vote for simply based on proxy labelling. However, models may be formed on underlying patterns which (despite lack of expected accuracy) hint towards more trends within the data.

Model Prediction

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest (1)	0.566	0.575	0.560	0.560	0.575	0.093
CN2 Rule Induction (1)	0.525	0.536	0.535	0.534	0.536	0.043
Logistic Regression (1)	0.572	0.571	0.550	0.553	0.571	0.077
Tree (1)	0.556	0.570	0.550	0.552	0.570	0.075
Neural Network (1)	0.552	0.554	0.552	0.551	0.554	0.077
Naive Bayes (1)	0.570	0.557	0.559	0.562	0.557	0.099

Figure 5.3: Varying model predictability.

Random Forest is the most accurate despite being only slightly better than a guess. This failure is also shown by the **Matthews Correlation Coefficient (MCC)** which measures how good models are for binary classifications (**Trump, Biden**). With all models $MCC < 0.1$ they are all very bad at determining the correct class.

While the models are weak and don't predict who someone votes for very accurately (likely due to the vast range of individuals within the sub-set who did not vote for Biden or Trump respectively but were classified as such); the model can be used to help suggest trends within the data that point to an overall classification.

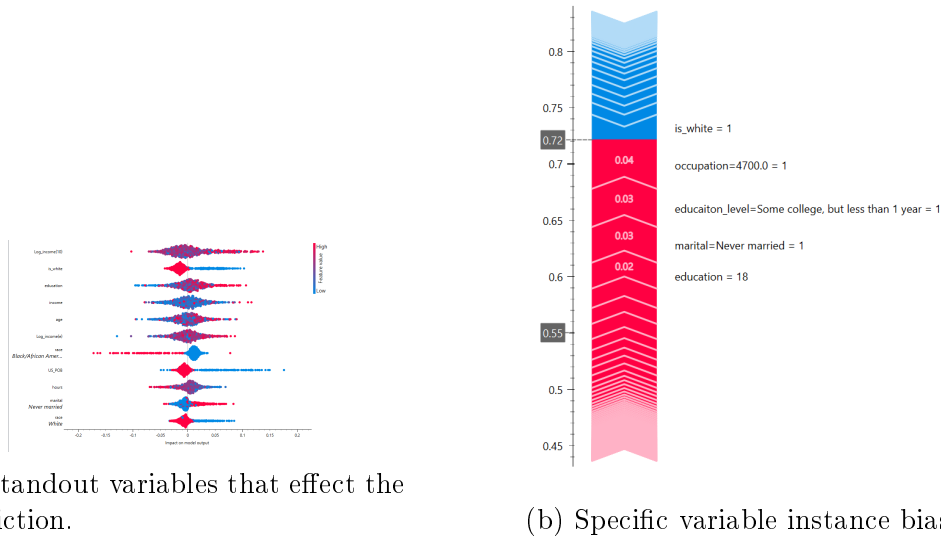


Figure 5.4: How variables affect the voting prediction.

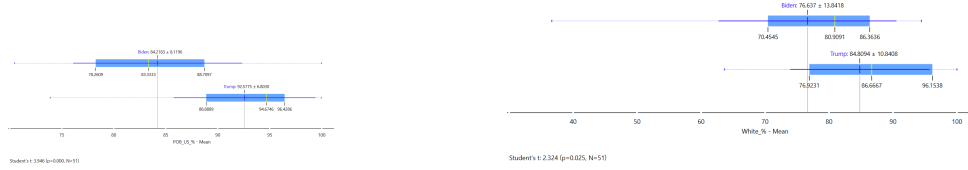
Figure 5.4a shows how varying values influence the models prediction towards someone voting for Biden, with factors such as:

- **Higher income** levels correlate with a stronger prediction for Biden.
- Identifying as **white** acts as a negative predictor for Biden, pushing the classification toward Trump.
- Specific **occupation codes** (such as 4700, representing Retail/Service roles) contribute positively to a Biden prediction.
- Being **never married** or single increases the probability of a Biden vote, aligning with more liberal urban demographics.

- Educational exposure, specifically **starting but not completing college**, is shown to increase the likelihood of a Biden prediction.

5.0.3 Are there any racial factors that effect voting habits?

Following the identification of **Race** on voting patterns, what is the variation between the candidates.



(a) Proportions of voters **born in the US** for each Candidate.

(b) Proportions of voters who are **White** for each Candidate.

Figure 5.5: Race and Background of each candidate's voters.

With $P = 0.000$ and $P = 0.025$ respectively (< 0.05), there is sufficient evidence to suggest that Trump-voting states ($\approx 84.8\%$) have a higher proportion of **white** voters than Biden-voting states ($\approx 76.6\%$) as well as Biden-voting states containing a significantly higher proportion of voters born **outside the US**.

5.1 Summary

In summary, predicting individual voting behaviour is not feasible from the data provided; however, features such as **Place of Birth**, **White or not**, and **Industry** can all help point towards a preference for one candidate over another. Conversely, for specific categories such as shipping office workers or widowed individuals, there is a lack of cases to support a clear classification. This essentially adds noise to the data. In this instance, utilising a larger sample size or individual-level voting records would allow for a more accurate model that can identify patterns more easily.

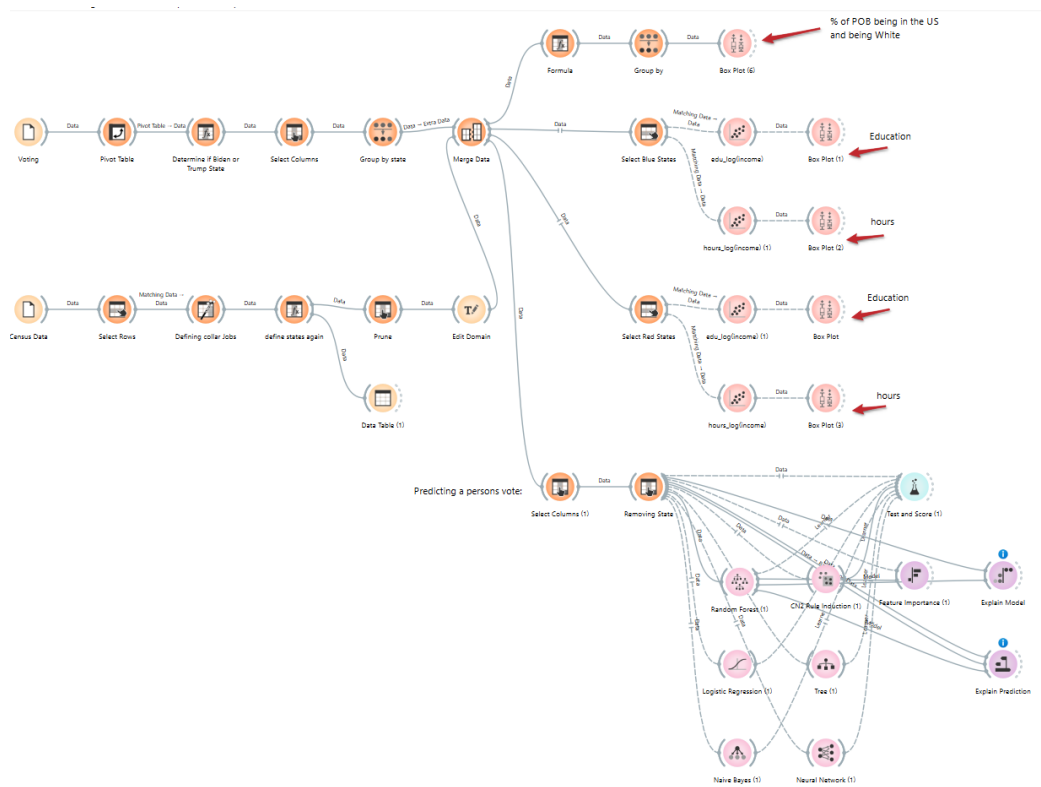


Figure 5.6: My Data mining Structure.